

L10. Principles of Quantum Error Correction

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Challenges

- Before 1995, it was believed that the concept of error correction could not apply to quantum systems, for the following reasons:
 - Errors are continuous.
 - Quantum states cannot be cloned.
 - Quantum states collapse when measured.
- It turns out there are ways around all of these objections.
 - Errors can be made orthogonal and distinguishable.
 - Entanglement also replaces the role played by redundant copies in classical error correction.
 - The part of the Hilbert space containing the quantum information to be preserved need not be measured; only the effect of the environment need be determined by a measurement.

Outline

- Representation of Quantum Errors
- 3-Bit Repetition Code
- Understanding Error Detection in the Stabilizer Formalism

9.1 Representation of Quantum Errors

- Interaction with the environment will transform and entangle the states of the qubit and its environment. In general, we expect the following process:

$$|e_0\rangle|0\rangle \rightarrow |e_1\rangle|0\rangle + |e_2\rangle|1\rangle,$$

$$|e_0\rangle|1\rangle \rightarrow |e_3\rangle|0\rangle + |e_4\rangle|1\rangle,$$

where $|e_{0-4}\rangle$ are the states of the environment.

- For practical quantum computing, we expect weak coupling between the qubit and its environment.

- Hence, we expect

$$\sqrt{|\langle e_0|e_1\rangle|^2} \approx \sqrt{|\langle e_0|e_4\rangle|^2} \approx 1,$$

$$\langle e_2|e_2\rangle, \langle e_3|e_3\rangle \ll 1.$$

- To discuss the interaction with the environment, it is useful to rewrite projection operators, which satisfy

$$P_0|\psi\rangle = |0\rangle\langle 0|\psi\rangle, \quad P_1|\psi\rangle = |1\rangle\langle 1|\psi\rangle,$$

as

$$P_\alpha = \frac{I + (-1)^\alpha Z}{2}.$$

- We can now write

$$|e_0\rangle|\psi\rangle \rightarrow [|e_1\rangle I + |e_2\rangle X] P_0|\psi\rangle + [|e_3\rangle X + |e_4\rangle I] P_1|\psi\rangle,$$

where $|e\rangle O|\psi\rangle \equiv |e\rangle \otimes O|\psi\rangle$.

- Plugging in the explicit form of P_α , we find

$$\begin{aligned} |e_0\rangle|\psi\rangle &\rightarrow \left[\frac{|e_1\rangle + |e_4\rangle}{2} I + \frac{|e_1\rangle - |e_4\rangle}{2} Z \right. \\ &\quad \left. + \frac{|e_3\rangle + |e_2\rangle}{2} X + \frac{|e_3\rangle - |e_2\rangle}{2} ZX \right] |\psi\rangle \\ &\equiv (|d\rangle I + |a\rangle X + |b\rangle Y + |c\rangle Z) |\psi\rangle. \end{aligned}$$

- The four operators I , Z , X , and Y correspond to no error, phase error, bit-flip error, and combined bit-flip and phase error, respectively.
 - We notice that

$$|d\rangle = \frac{|e_1\rangle + |e_4\rangle}{2} \approx |e_0\rangle,$$

but $|a\rangle$, $|b\rangle$, and $|c\rangle$, which are associated with various errors, are ket-vectors with small amplitudes. This is consistent with our weak-coupling assumption.

- For one qubit, there are three terms of errors.

- When we generalize the result to 1-bit errors for n -bit codewords $|\Psi_n\rangle$, we can write

$$|e_0\rangle|\Psi_n\rangle \rightarrow |d\rangle|\Psi_n\rangle + \sum_{i=1}^n (|a_i\rangle X_i |\Psi_n\rangle + |b_i\rangle Y_i |\Psi_n\rangle + |c_i\rangle Z_i |\Psi_n\rangle).$$

- Altogether, there are $(3n + 1)$ terms.

9.2 3-Bit Repetition Code

Take-Home Messages

- While measurement can alter a state, there exist less obvious forms of measurement for error detection. In a repetition code, one can use CNOT gates to extract the information without directly measuring the qubits.
- Errors, in general, leads to information leakage into the non-encoding space. One can correct the errors by returning to the encoding space without revealing any information of the logical qubit.

Error Correction

- A classical code is defined as being a set of bit strings with certain properties.
 - For example, the set of bit strings $\{000, 111\}$ is a well-known code. One can send each of the three bits through a communication channel, in which, with probability p , a bit is flipped. Decoding can be done by computing the majority vote of the bits received.
 - With no coding (sending a single-bit message), the probability of error is p . Using the triple redundancy code, the probability of the decoded bit being wrong is $O(p^2)$.
 - To leading order, the error probability now depends quadratically instead of linearly on p ; this is the key signature of error correction in action.

- To encode each logic qubit in multiple physical qubit, we replace $|0\rangle$ and $|1\rangle$ by the codewords, e.g.,

$$|\tilde{0}\rangle = |0\rangle|0\rangle|0\rangle = |000\rangle,$$

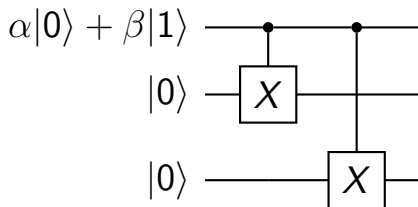
$$|\tilde{1}\rangle = |1\rangle|1\rangle|1\rangle = |111\rangle.$$

- One bit flip moves the states to the non-encoding space $\{|001\rangle, |010\rangle, |100\rangle, |110\rangle, |101\rangle, |011\rangle\}$, therefore can be detected.
- But as measurement can alter a state, one must come up with less obvious form of measurement.

The 3-Bit Repetition Code

- Preparation

$$|\psi\rangle \equiv \alpha|000\rangle + \beta|111\rangle.$$



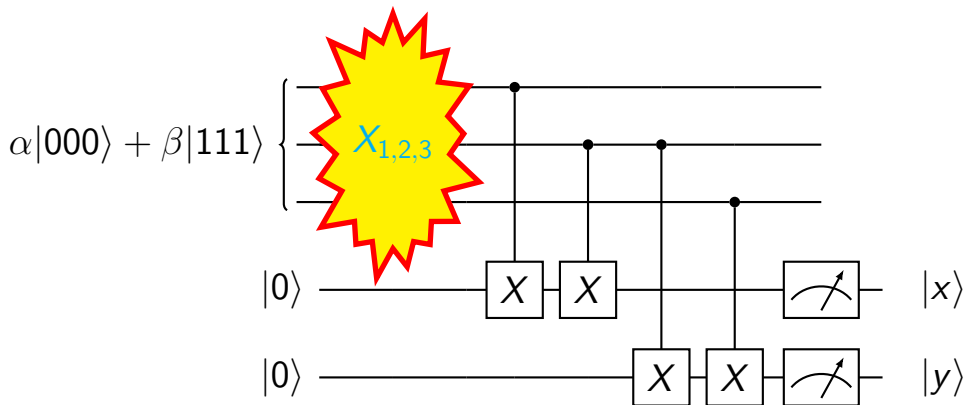
- Occasionally, a bit is flipped (with a small probability p).

$$|\psi_1\rangle = X_1|\psi\rangle = \alpha|100\rangle + \beta|011\rangle,$$

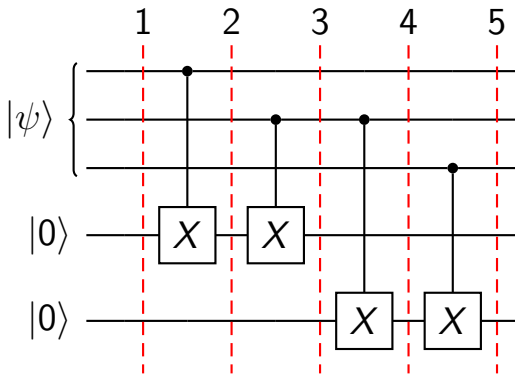
$$|\psi_2\rangle = X_2|\psi\rangle = \alpha|010\rangle + \beta|101\rangle,$$

$$|\psi_3\rangle = X_3|\psi\rangle = \alpha|001\rangle + \beta|110\rangle.$$

- Detection: We cannot measure the three qubits, but can use CNOT gates to extract the information.



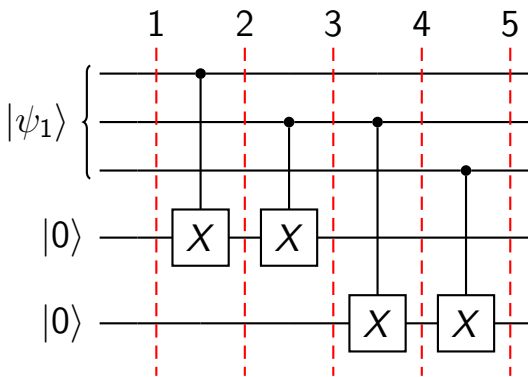
• Case 1: No error.



- 1: $(\alpha|000\rangle + \beta|111\rangle)|00\rangle$
 2: $(\alpha|0000\rangle + \beta|1111\rangle)|0\rangle$
 3: $(\alpha|000\rangle + \beta|111\rangle)|00\rangle$
 4: $\alpha|00000\rangle + \beta|11101\rangle$
 5: $(\alpha|000\rangle + \beta|111\rangle)|00\rangle$

$$|\psi\rangle = I|\psi\rangle \rightarrow |x\rangle = |0\rangle, \quad |y\rangle = |0\rangle$$

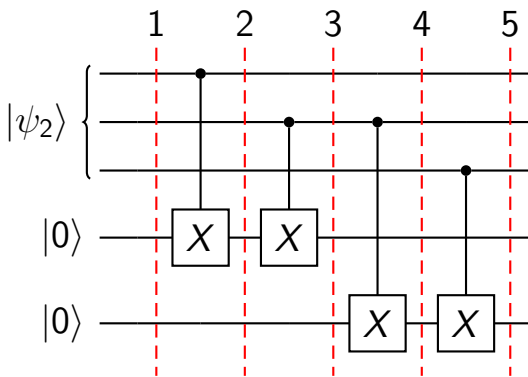
• Case 2: X_1 error.



- 1: $(\alpha|100\rangle + \beta|011\rangle)|00\rangle$
 2: $(\alpha|1001\rangle + \beta|0110\rangle)|0\rangle$
 3: $(\alpha|1001\rangle + \beta|0111\rangle)|0\rangle$
 4: $\alpha|10010\rangle + \beta|01111\rangle$
 5: $(\alpha|100\rangle + \beta|011\rangle)|10\rangle$

$$|\psi_1\rangle = X_1|\psi\rangle \rightarrow |x\rangle = |1\rangle, \quad |y\rangle = |0\rangle$$

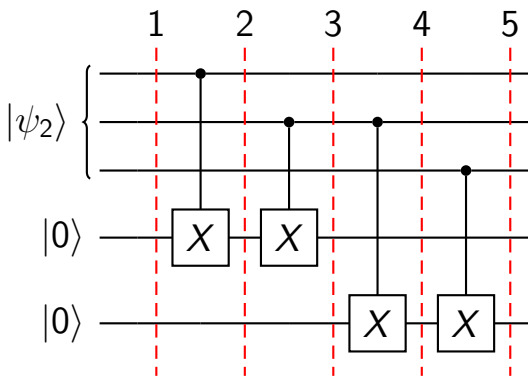
• Case 3: X_2 error.



$$\begin{aligned} 1: & (\alpha|010\rangle + \beta|101\rangle)|00\rangle \\ 2: & (\alpha|0100\rangle + \beta|1011\rangle)|0\rangle \\ 3: & (\alpha|0101\rangle + \beta|1011\rangle)|0\rangle \\ 4: & \alpha|01011\rangle + \beta|10110\rangle \\ 5: & (\alpha|010\rangle + \beta|101\rangle)|11\rangle \end{aligned}$$

$$|\psi_1\rangle = X_2|\psi\rangle \rightarrow |x\rangle = |1\rangle, \quad |y\rangle = |1\rangle$$

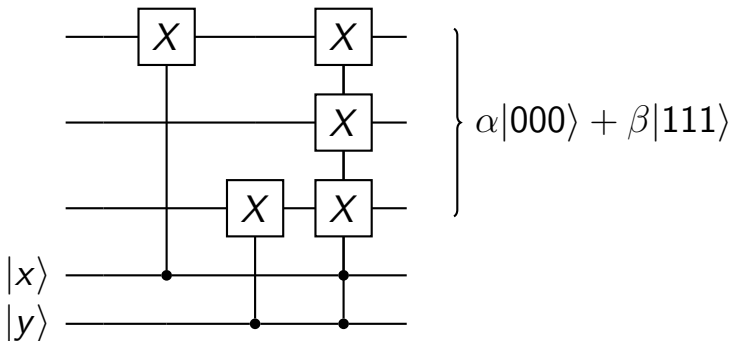
• Case 4: X_3 error.



$$\begin{aligned}
 1: & (\alpha|001\rangle + \beta|110\rangle)|00\rangle \\
 2: & (\alpha|0010\rangle + \beta|1101\rangle)|0\rangle \\
 3: & (\alpha|0010\rangle + \beta|1100\rangle)|0\rangle \\
 4: & \alpha|00100\rangle + \beta|11001\rangle \\
 5: & (\alpha|001\rangle + \beta|110\rangle)|01\rangle
 \end{aligned}$$

$$|\psi_1\rangle = X_3|\psi\rangle \rightarrow |x\rangle = |0\rangle, \quad |y\rangle = |1\rangle$$

- In addition to detection, we can directly correct the error by the following circuit without measuring.



- The essence is that we can restore $|\psi\rangle$ without revealing any information about α and β .

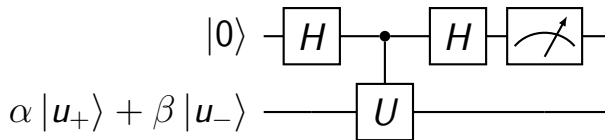
- In addition to bit-flip errors, there can also be phase errors, e.g.,

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}} \rightarrow \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

- A generalization of the 3-bit coding to 9-bit can correct both bit-flip and phase errors. See, e.g., D. Gottesman, arXiv:0904.2557, Sec. 2.1.

9.3 Understanding Error Detection

- To understand the connection between error detection and operator measurement, we digress to the following circuit for measurement of an operator U acting on a single qubit.



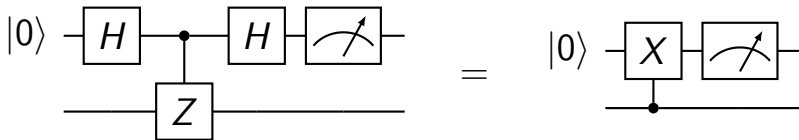
- Here, we suppose U has eigenvalues ± 1 and corresponding eigenvector $|u_{\pm}\rangle$. So a single-qubit state $|\psi\rangle$ can be generically expressed as $\alpha |u_+\rangle + \beta |u_-\rangle$.

- Following the circuit from left to right, the two-qubit state evolves as

$$\begin{aligned}
 & |0\rangle (\alpha |u_+\rangle + \beta |u_-\rangle) \\
 \xrightarrow{H} & \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) (\alpha |u_+\rangle + \beta |u_-\rangle) \\
 \xrightarrow{\text{Ctrl-}U} & \frac{|0\rangle (\alpha |u_+\rangle + \beta |u_-\rangle)}{\sqrt{2}} + \frac{|1\rangle (\alpha |u_+\rangle - \beta |u_-\rangle)}{\sqrt{2}} \\
 \xrightarrow{H} & \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \left(\frac{\alpha |u_+\rangle + \beta |u_-\rangle}{\sqrt{2}} \right) + \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \left(\frac{\alpha |u_+\rangle - \beta |u_-\rangle}{\sqrt{2}} \right) \\
 = & \alpha |0\rangle |u_+\rangle + \beta |1\rangle |u_-\rangle.
 \end{aligned}$$

- When the measurement result is 0, the circuit outputs $|u_+\rangle$. Otherwise, it outputs $|u_-\rangle$.

- As an example of the circuit that can be used to measure a single-qubit operator U , we consider $U \equiv Z$.



- This implies that we can alternatively interpret the repetition code as measuring the product of Z -gates operating on the encoded qubits.
- For that reason, let us define

$$U_1 = Z_1 Z_2, \quad U_2 = Z_2 Z_3.$$

- Note that the two operators leave the logical qubit state

$$|\psi\rangle \equiv \alpha|000\rangle + \beta|111\rangle$$

unchanged,

$$U_1 |\psi\rangle = |\psi\rangle, \quad U_2 |\psi\rangle = |\psi\rangle,$$

and are, thus, said to *stabilize* the state. In other words, U_1 and U_2 stabilize all states in the codespace $\mathcal{C} = \text{span}\{|000\rangle, |111\rangle\}$.

- Alternatively, $\mathcal{C}$ is in the $+1$ -eigenstate space of U_1 and in that of U_2 .

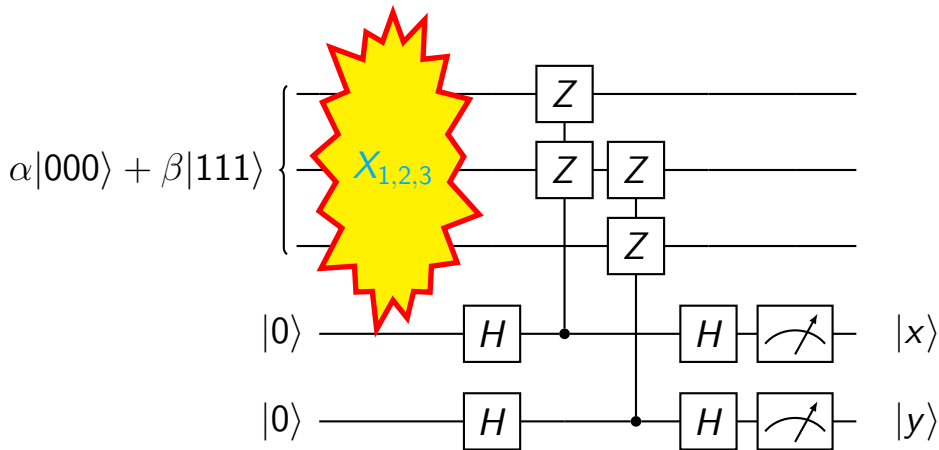
- The orthogonal subspace

$$\mathcal{F} = \text{span}\{|001\rangle, |010\rangle, |100\rangle, |011\rangle, |101\rangle, |110\rangle\}$$

is the non-coding space. To detect a single bit flip, we need to determine whether a state is in $\mathcal{C}$ or $\mathcal{F}$.

- For that purpose, we can perform projective measurement of U_i ($i = 1, 2$) using a controlled- U_i gate with a corresponding ancilla qubit, as described earlier. Note that we are not measuring Z_0 , Z_1 , and Z_2 and then multiplying the results.

- An equivalent circuit for error detection, which incorporates U_1 and U_2 explicitly, can be constructed.



- The operators U_1 and U_2 each has two distinct eigenvalues ± 1 . The measurement of the operators, for zero and one-qubit errors, yields

	I	X_1	X_2	X_3
$U_1 = Z_1 Z_2$	+1	-1	-1	+1
$U_2 = Z_2 Z_3$	+1	+1	-1	-1

- The measurement of U_i distinguishes between the error state and the non-error state. The results contain the identical information as the measurement of $|x\rangle$ and $|y\rangle$, discussed in the original circuit. They are known as the **syndrome**, which can be used to identify and locate errors.

- Whenever the stabilizer U_i anti-commutes with an error operator X_j , the measurement of U_i after the error operator will yield an eigenvalue -1 :

$$U_i X_j |\psi\rangle = -X_j U_i |\psi\rangle = -X_j |\psi\rangle .$$

- As in the present case, one chooses stabilizers to be tensor products of Pauli operators, then any error that is also a combination of Pauli operators will either commute or anti-commute with the stabilizers. If all error operators anti-commute with distinct sets of stabilizers, then we can perform error detection.

- To generalize the idea, one can define a set of m operator $\{U_i\}$, which commute with each other and have distinct eigenvalues ± 1 . One then encodes with n qubits in the $+1$ -eigenvalue sector of $\{U_i\}$ (with 2^{n-m} basis vectors).
- Without directly measuring the encoded qubits, one can identify errors caused by the environment, and correct errors accordingly.
- This idea leads to the most important family of quantum codes, known as the stabilizer codes, discovered and defined by Gottesman.

- In our case, the 3-bit code is known as a $[3, 1]$ ($n = 3, k = n - m = 1$) stabilizer code with U_1 and U_2 being stabilizers.

State	U_1	U_2	Probability
$\alpha 000\rangle + \beta 111\rangle$	+1	+1	$(1 - p)^3$
$\alpha 001\rangle + \beta 110\rangle$	+1	-1	$p(1 - p)^2$
$\alpha 010\rangle + \beta 101\rangle$	-1	-1	$p(1 - p)^2$
$\alpha 100\rangle + \beta 011\rangle$	-1	+1	$p(1 - p)^2$

- For more information, see, e.g., M. A. Nielsen and I. L. Chuang, Quantum Computation and Quantum Information, 10th Anniversary ed. (Cambridge, 2011), Chap. 10.

Summary

- Phase error, bit-flip error, and combined bit-flip and phase error for a single qubit can be represented by Z , X , and Y gates, respectively.
- Redundancy in coding with CNOT gates for error detection can lead to error correction without directly measuring the qubits.

References

- M. A. Nielsen and I. L. Chuang, Quantum Computation and Quantum Information, 10th Anniversary ed. (Cambridge, 2011), Chap. 10.
- I. L. Chuang, Quantum Error Correction, in Quantum Machines: Measurement and Control in Engineered Quantum Systems, edited by M. Devoret, B. Huard, R. Schoelkopf, and L. F. Cugliandolo, Oxford University Press (2014).